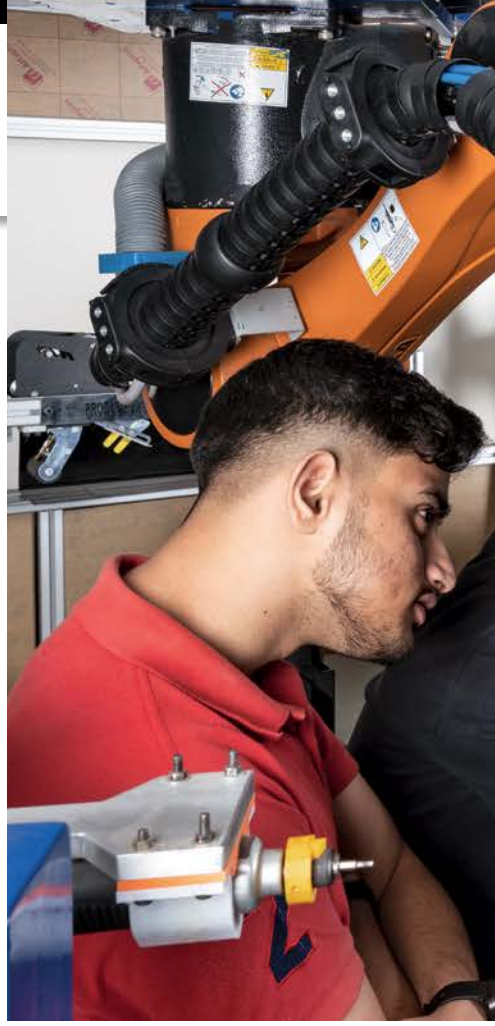
THE STEM OF PROGRESS

HIGHER INVESTMENT IN STEM EDUCATION WILL BE CRUCIAL FOR EMPLOYMENT GENERATION AND GROWTH IN A TECH-DRIVEN UNIVERSE



HARDIK CHHABRA

In a world with constant churning in technological innovation, demand for manpower in Science, Technology, Engineering and Mathematics (STEM) has been rapidly growing. Advancements in Robotics, Artificial Intelligence (AI), Machine Learning (ML), cloud computing, and the Internet of Things (IoT) have accelerated demand for a talent reservoir around STEM. According to the Future of Jobs Report 2023 by the World Economic Forum, over 75 per cent of firms are looking to adopt AI, big data analytics and cloud computing in the next five years. Sectors like healthcare and financial services demand a convergence of STEM subjects. STEM fields will be critical in research that addresses the impact of climate change and future pandemics and to ensure food security.

WHY IT IS A GAME CHANGER

STEM education is crucial for developing nations to keep pace with tech advancements and remain globally competitive. By sharpening focus on this broad swathe of specialised but interrelated knowledge domains, countries can strengthen their research capabilities, build new tech and offer local solutions to problems. This reduces reliance on import of technology and solutions, while enabling genuine knowledge exchange.

STEM education is crucial for employment generation too. Demand for STEM jobs in India rose by 44 per cent between 2016 and 2021 and is expected to grow by 50 per cent by 2025. The Future of Jobs Report, 2023, indicates that the top emerging jobs in India will be in AI and ML. With a population of 1.4 billion and a median age of 27, India must train its youth to exploit the full potential of this demographic dividend. That demands more budgetary allocation and higher focus on STEM education.

WHAT INDIA NEEDS TO MASTER

STEM education will play a pivotal role as India aspires to gradually move from being a services-led economy to a manufacturing-based one. With 34 per cent of its graduates being in STEM subjects, India is

THE CHANGE-MAKERS

10 MILLION

Projected number of Indian STEM graduates in 2025

43%

Of all STEM graduates in India are women, higher than the US (34 per cent) and the UK (38 per cent)